SPINE · NEET-SS / FET / INI-SS

Thoracolumbar trauma

1. Anatomy and biomechanics

- The **thoracolumbar junction (T11-L2)** is the commonest site of spinal fractures: transition from the rigid, rib-supported, kyphotic thoracic spine (coronal facets) to the mobile, lordotic lumbar spine (sagittal facets).
- The **thoracic canal is narrow** (and the cord blood supply tenuous: watershed **T4-T9**; **artery of Adamkiewicz** usually left-sided **T9-L2**), so thoracic injuries cause more **complete cord injuries**. The **cord ends (conus) at L1-L2**; below this, injuries affect the **cauda equina** (better prognosis).
- **Denis three-column concept: anterior** (ALL, anterior two-thirds of the body and disc), **middle** (posterior third of body and disc, **PLL**), **posterior** (pedicles, facets, laminae, **posterior ligamentous complex: supraspinous, interspinous ligaments, ligamentum flavum, facet capsules**). **Instability when two or more columns fail** (middle column key in Denis's model).
- The **posterior ligamentous complex (PLC)** is the posterior **tension band**; its integrity is the main determinant of stability in modern classifications. MRI: **STIR hyperintensity** in interspinous region, disrupted black line of the supraspinous ligament/ligamentum flavum; radiographs/CT: **widened interspinous distance**, facet diastasis/perch, **splaying of spinous processes**.

2. Classifications

Denis (1983)

- **Compression** (anterior column only; stable), **burst** (anterior + middle; types A-E), **flexion-distraction (seat-belt/Chance)**,

fracture-dislocation (all three columns).

TLICS (Thoracolumbar Injury Classification and Severity score, Vaccaro)

Component	Points
Morphology: compression 1, **burst 2**, **translation/rotation 3**, **distraction 4**	1-4
Neurology: intact 0; **nerve root 2**; **complete cord/conus 2**; **incomplete cord/conus 3**; **cauda equina 3**	0-3
PLC: intact 0; **suspected/indeterminate 2**; **injured 3**	0-3

- **Under 4: non-operative; 4: either; over 4 (5 or more): operative.**

AO Spine thoracolumbar classification (2013)

- **Type A (compression):** A0 minor/process fractures; **A1 wedge** (one endplate); **A2 split/pincer** (both endplates, coronal split); **A3 incomplete burst** (one endplate + posterior wall); **A4 complete burst** (both endplates + posterior wall).
- **Type B (tension band failure): B1 transosseous posterior tension band disruption (bony Chance)**; **B2 posterior ligamentous tension band disruption** (with any A injury of the body); **B3 hyperextension** (anterior tension band disruption: ankylosed spines).
- **Type C (displacement/translation in any direction):** all elements fail; **the most unstable**.
- **Neurology modifiers:** N0 intact, N1 transient, N2 radicular, N3 incomplete cord/cauda, N4 complete cord, NX unexaminable; **case modifiers M1 (indeterminate PLC injury), M2 (patient-specific comorbidity: ankylosing spondylitis, osteoporosis, burns)**.
- **TL AOSIS** score guides treatment (similar to TLICS).

3. Specific injuries and their management

Compression (wedge) fractures

- Anterior column only; **stable**. **Non-operative** (brace or early mobilisation as pain allows). Concern if **loss of anterior height over 50%**, **kyphosis over 20-30 degrees**, or multiple contiguous fractures (consider PLC injury).
- **Osteoporotic vertebral compression fractures:** commonest osteoporotic fracture; often asymptomatic; **MRI STIR** shows acute oedema; treat pain, bone health (anti-osteoporotic therapy; **teriparatide** speeds healing). **Vertebroplasty/kyphoplasty** for persistent severe pain (evidence mixed: **sham-controlled trials (Buchbinder, Kallmes) showed no benefit over sham**; some trials (VERTOS IV/ II) differ). **Kümmell disease:** delayed post-traumatic vertebral collapse with an **intravertebral vacuum cleft** (AVN).

Burst fractures

- **Axial load**; anterior + middle column failure with **retropulsion** of the posterior wall into the canal; **interpedicular widening** on AP radiograph.
- **Neurologically intact burst fracture with intact PLC (TLICS 2-3, AO A3/A4 N0): non-operative treatment** (early mobilisation with or without a TLSO brace) gives **similar outcomes to surgery** (**Wood et al.**, RCT: no difference at long-term follow-up; **brace vs no brace (Bailey 2014)**: no difference). **Canal compromise percentage and kyphosis alone do not dictate surgery**; retropulsed fragments remodel (resorption of canal fragments).
- **Surgical indications: neurological deficit with canal compromise**, **PLC disruption**, progressive kyphosis/neurology, **kyphosis over 25-30 degrees** or **height loss over 50%** (relative, debated), polytrauma requiring mobilisation.

- **Surgery: posterior pedicle screw fixation** (short-segment: one level above and below ± screws in the fractured vertebra (**intermediate screws**) improve kyphosis correction; long segment for unstable/osteoporotic), **indirect decompression by ligamentotaxis** (works only if the **PLL is intact** and within about 4 days), **direct decompression** (posterolateral transpedicular or anterior corpectomy) for significant canal compromise with deficit; **anterior corpectomy and reconstruction** for severe comminution (**load-sharing classification, McCormack: score 7 or more** = anterior column reconstruction needed; points for comminution, apposition, kyphosis correction).

Flexion-distraction (Chance) injuries

- **Lap belt** (axis of flexion anterior to the spine): **tension failure of all three columns**, horizontal fracture through spinous process, lamina, pedicles and body (**bony Chance**, B1) or through ligaments/disc (**ligamentous**, B2).
- **Associated intra-abdominal injuries in about 40-50%** (bowel perforation, mesenteric tears): **"seat-belt sign"** on the abdomen: CT abdomen, surgical review.
- **Treatment: purely bony Chance fracture** can heal in a **hyperextension cast/brace** (if neurologically intact and reduced); **ligamentous or disc injuries do not heal** reliably: **posterior compression instrumentation and fusion**.

Fracture-dislocations (AO C)

- Shear, rotation or translation; **highest rate of neurological injury** (complete cord injury common); **posterior instrumentation (long-segment) with decompression and fusion** in all.

Hyperextension injuries in ankylosed spines (AS, DISH)

- **Long-lever fractures through the fused spine**, highly unstable, frequently missed (**image the whole spine with CT ± MRI**), **epidural**

haematoma with delayed neurological deterioration. **Immobilise in the pre-injury (kyphotic) position**; **long posterior fixation (at least 2-3 levels above and below)**. High mortality in the elderly.

4. Sacrum and lumbosacral junction (overlap with pelvis)

- **Spinopelvic dissociation (U/H-type sacral fractures):** falls from height/suicidal jumpers ("**jumper's fracture**"); bilateral vertical sacral fractures with a transverse component; cauda equina injury; **lumbopelvic (triangular) fixation** ± decompression.
- **Roy-Camille** classification of transverse sacral fractures.

5. Neurological injury management (see cervical notes)

- **ASIA examination**; spinal shock (bulbocavernosus reflex); maintain **MAP 85-90 mmHg** for about 7 days; **early decompression (within 24 hours)** where feasible; **no routine steroids**.
- **Conus medullaris syndrome** (T12-L1): mixed UMN and LMN signs, early bowel/bladder involvement, saddle anaesthesia; **cauda equina syndrome** (below L1-L2): LMN pattern, better recovery.
- Rehabilitation: bladder (intermittent catheterisation), bowel programmes, pressure care, DVT prophylaxis (high VTE risk: LMWH within 72 hours), heterotopic ossification (hips), autonomic dysreflexia (above T6), spasticity, psychological care.

6. Penetrating spinal injury

- **Gunshot wounds:** steroids are **contraindicated** (more complications); **surgery rarely improves neurology**; remove bullets only for **progressive deficit**, **CSF leak**, **bullet within the canal at T12-L5 (cauda equina: possible benefit)**, **lead toxicity** risk if in the disc/CSF, or infection; **transcolonic injuries need broad-spectrum antibiotics** (debridement debated).

7. Paediatric thoracolumbar injuries

- **Chance fractures with lap belts** (with abdominal injuries), **apophyseal ring fractures** (slipped vertebral apophysis: adolescent; behaves like a disc herniation; may need excision), **SCIWORA** (commoner in younger children), **non-accidental injury** (multiple compression fractures). Post-traumatic **scoliosis/kyphosis** after SCI in young children (almost universal before puberty).

8. Complications

- **Post-traumatic kyphosis** (painful; corrective osteotomy: **pedicle subtraction osteotomy** about 30-40 degrees, Smith-Petersen about 10 degrees per level), **non-union/implant failure** (short-segment fixation without anterior support in comminuted fractures), **post-traumatic syrinx** (late neurological deterioration: MRI), **infection**, **dural tears**, **adjacent segment degeneration**, VTE, pressure sores.

Exam pearl: **TLICS: morphology (compression 1, burst 2, translation 3, distraction 4) + neurology + PLC; over 4 = surgery. Neurologically intact burst fracture with intact PLC = non-operative (Wood RCT), with or without a brace.**

Exam pearl: **Chance = lap belt flexion-distraction; 40-50% have abdominal visceral injury; bony Chance can heal in extension cast, ligamentous needs fusion. Ligamentotaxis needs an intact PLL. Load-sharing score 7 or more = anterior reconstruction.**

Exam pearl: **AO B = tension band failure (B1 bony Chance, B2 ligamentous, B3 hyperextension); AO C = translation (most unstable).** Ankylosed spine fractures: whole-spine imaging, long posterior fixation.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.